Chapter 4- BIOS, Booting, Partitioning & Recovery

🔷 1. What is BIOS? (Basic Input Output System)

👉 **BIOS** is a firmware (software on motherboard) that controls hardware during startup.

- You can see system info using **msinfo32** in Windows.
- BIOS is stored on the **motherboard chip**, not inside Windows.

⚙️ Main Functions of BIOS

- Controls **input/output devices** (keyboard, mouse, disk)
- Starts the **boot process**
- Checks hardware health

🔍 Daily-life Example

Like a **school principal** who checks everything before classes start.

🔷 2. Types of BIOS

🟤 1. Legacy BIOS

- Introduced around **2000**
- Supports max **2TB hard disk**
- Works with **MBR partition**
- No mouse support (keyboard only)

🔵 2. UEFI (Unified Extensible Firmware Interface)

- Modern BIOS (replaced legacy)
- Supports **more than 2TB disks**
- Supports **GPT partition**

- Has **mouse + keyboard support**
- Faster and more secure

🔍 **Daily-life Example**

- Legacy = Old keypad phone 📞
- UEFI = Modern smartphone 📱

🔷 3. BIOS Boot Process (Important Flow)

When system starts:

1. BIOS runs **POST (Power-On Self Test)** 🧪
2. Finds **boot device** (HDD/SSD/USB)
3. Loads bootloader:
    - Windows → **bootmgr**
    - Linux → **GRUB loader**
4. Bootloader loads OS into RAM
5. Login screen appears

⚠️ If boot files are deleted →
👉 ❌ **Error: "No Bootable Device Found"**

🔷 4. Hard Disk Partitioning

⭐ **What is Partitioning?**

Dividing a physical disk into multiple parts:

- C: → OS
- D: → Personal files
- E: → Games/work

⚙️ **Open Disk Management**

👉 Windows + R → diskmgmt.msc

💡 **IT Rule**

Counting starts from **0**
👉 First disk = **Disk 0**

🔍 **Daily-life Example**

Like dividing a **cupboard into shelves**

🔷 5. Partition Types: MBR vs GPT

🟤 **MBR (Master Boot Record)**

- Introduced in **1983**
- Max disk support: **2TB**
- Allows:
    - 3 Primary partitions
    - 1 Extended partition
    - Inside extended → Logical partitions

📍 **Storage Location**

Stored in **first sector of disk**

⚠️ **Disadvantage**

- If first sector is corrupted → data loss
- Recovery success ~70%

🔍 **Boot Identification**

➡️ Straight loading line = MBR

🧪 **Example**

Like storing all data in **one notebook**

🔵 **GPT (GUID Partition Table)**

- Modern and advanced system

✅ **Advantages:**

- Supports **>2TB disks**
- Up to **128 partitions**
- No need for extended/logical partitions
- Multiple backup copies
- Safer & faster recovery

🔍 **Boot Identification**

➡️ Spinning circle = GPT

🧪 **Example**

Like saving files with **multiple backups**

🔷 6. Partition Types (Important for IT Support)

🔹 **1. Primary Partition**

- Main partition for OS & data
- Required for OS installation

👉 Example: **C: Drive**

🔹 **2. Extended Partition (MBR only)**

- Works as a container
- Only **1 allowed**

🔹 **3. Logical Partition**

- Inside extended partition
- Cannot boot OS

👉 Example: D:, E:, F:

🔹 **4. Active Partition**

- Contains boot info
- Only **1 active partition allowed**

👉 Example: System Reserved

🔹 **5. System Partition**

- Contains boot files:
  - bootmgr
  - BCD
  - EFI folder

👉 Loads OS into RAM

🔷 **7. EFI System Partition (ESP)**

👉 Used only in **UEFI systems**

📁 **Contains:**

- bootmgr.efi
- EFI boot files

- Drivers

⚙️ **Purpose:**

- Controls boot process
- Loads login screen

⚠️ Without ESP → System won't boot

🔷 8. Recovery Partition

👉 Hidden partition created by Windows

🛠️ **Uses:**

- System repair
- Reset PC
- Troubleshooting
- Backup restore

📦 **Contains:**

- Windows Recovery Environment (WinRE)

🔍 **Example**

Like a **first-aid kit** 🩹 **for your PC**

🔷 9. Entering Recovery Mode

✅ **Method 1 (Settings)**

Settings → System → Recovery → Advanced Startup

⚠️ **Method 2 (Force Shutdown)**

- Restart system
- Force shutdown 5–6 times
- Auto recovery starts

🔷 10. Boot Flow (Full Understanding)

1. BIOS/UEFI starts
2. System partition loads boot files
3. Boot manager loads OS
4. Login screen appears
5. If OS fails → Recovery starts

🔷 11. BIOS vs UEFI Check

🟦 **Method 1: Boot Screen**

- Straight line → Legacy
- Spinning circle → UEFI

🟩 **Method 2: BIOS Interface**

- Mouse works → UEFI
- Only keyboard → Legacy

🔷 12. BIOS & Boot Keys (Important for Support)

Brand	BIOS Key	Boot Menu
Acer	F2 / DEL	F12 / ESC
Dell	F2	F12
HP	F10	ESC
Lenovo	F1 / F2	F12
ASUS	F2 / DEL	F8

👉 Used during troubleshooting (OS install, boot issues)

🔷 13. BIOS Security Settings (HP Example)

🔐 **BIOS Administrator Password**

- Protects BIOS access
- Prevents unauthorized changes

🔷 14. Secure Boot (VERY IMPORTANT) 🔥

👉 Available only in **UEFI**

⚙️ **What it does:**

- Protects system from malware during boot
- Checks files before OS loads

💡 **Why needed?**

Problem:

- Malware loads into RAM **before antivirus starts**

Solution:

- Secure Boot scans during BIOS stage

🔍 **Flow:**

BIOS → Bootmgr → OS → Antivirus
👉 Secure Boot protects **before OS loads**

🧪 **Example**

Like checking bags **before entering airport** ✈️

🔷 15. BIOS Advanced Boot Options

🔌 USB Boot Issue (Common Problem)

👉 Problem:
Pendrive not detecting

👉 Solution:

- Go to BIOS → Boot Options
- Enable **USB Boot**

🔢 Boot Order Priority

Example:

1. SSD
2. USB
3. Network Boot

💻 Types of SSD

- SATA SSD
- NVMe SSD
- M.2 SSD

🌐 Network Boot (Advanced Concept)

- Used in companies/malls
- OS loads via **Ethernet from server**
- No hard disk needed

🔷 16. Windows 11 ISO Download

1. Go to Microsoft website
2. Select:
   - Windows 11 ISO (x64)
   - Multi-edition
3. Choose language
4. Download 64-bit ISO

🔷 17. Rufus Tool

👉 Used to create **bootable USB**

- Download from official site
- File: rufus-x.x.exe

🎯 IT Support Practical Tips (VERY IMPORTANT)

✔ Always check:

- Boot order
- Partition type (MBR/GPT)
- BIOS mode (UEFI/Legacy)
- Secure Boot status

✔ Common errors:

- No boot device → boot files missing
- USB not detecting → USB boot disabled
- OS not installing → wrong partition type

✔ Fast troubleshooting mindset:

- Identify problem quickly

- Check BIOS first
- Then disk & bootloader
- Then OS

🧠 **Final Understanding**

👉 BIOS = Starts system
👉 Bootloader = Loads OS
👉 Partition = Organizes disk
👉 Secure Boot = Protects startup
👉 Recovery = Fixes system

📘 Chapter 4 - SUMMARY

🧠 BIOS (Basic Input Output System)

👉 BIOS is firmware stored on motherboard chip
👉 It starts the system and checks hardware

⚙️ **Functions:**

- Hardware check (POST) 🧪
- Start boot process 🔄
- Manage input/output devices ⌨️

🔵 Legacy vs UEFI

Feature	Legacy BIOS 🟤	UEFI 🔵
Disk support	Up to 2TB	More than 2TB
Partition	MBR	GPT
Interface	Keyboard only	Mouse + GUI
Speed	Slow	Fast

👉 UEFI = Modern & secure 🚀

🔄 Booting Process

👉 Step-by-step:

1. Power ON ⚡
2. BIOS/UEFI starts 🧠
3. POST check 🧪
4. Boot device detected 💾

5. Bootloader loads OS
6. Login screen appears

💾 Partitioning

👉 Disk ko parts me divide karna:

- C: → OS
- D: → Data

💡 Example: Cupboard shelves

🔵 MBR vs GPT (VERY IMPORTANT)

🟤 MBR

- Max 2TB
- 4 partitions
- Risky (single copy)

🔵 GPT

- 2TB
- 128 partitions
- Backup available

👉 GPT is safer & modern ✅

🔹 Partition Types

- Primary → OS install
- Extended → Container
- Logical → Inside extended
- Active → Boot partition
- System → Boot files

🔷 EFI Partition (UEFI)

👉 Contains:

- bootmgr.efi
- Boot files

👉 Without EFI → System will NOT boot ❌

🛠️ Recovery Partition

👉 Hidden partition for:

- Reset PC
- Troubleshooting
- Repair

💡 Like first aid kit 🩹

🔐 Secure Boot (VERY IMPORTANT)

👉 Works in UEFI only

👉 Function:

- Blocks malware during boot
- Verifies files before OS loads

🔧 Boot Issues (Common)

❌ No Boot Device:

- Wrong boot order
- OS missing
- Disk failure

❌ USB Not Detect:

- USB boot disabled

🌐 **Advanced Concepts**

👉 Network Boot:

- OS loads from server

👉 Rufus:

- Creates bootable USB

🎯 **CONCLUSION**

👉 Full boot flow:
BIOS → Bootloader → OS → Login

👉 Core understanding:

- BIOS = Start system
- Partition = Organize disk
- Bootloader = Load OS
- Recovery = Fix issues

👉 Final line:
"If you understand booting deeply, you can fix 80% system problems" 🔥

💼 🔥 Q & A

🔹 BIOS QUESTIONS

❓ Q1: What is BIOS?

👉 **Answer:**
BIOS is firmware stored on the motherboard that initializes hardware and starts the booting process.
It acts as a bridge between hardware and operating system.

❓ Q2: What is POST?

👉 **Answer:**
POST (Power-On Self Test) is a process where BIOS checks hardware components like RAM, CPU, and storage before booting.

❓ Q3: Where is BIOS stored?

👉 **Answer:**
BIOS is stored in a chip on the motherboard, not in the operating system.

🔹 UEFI vs LEGACY QUESTIONS

❓ Q4: Difference between BIOS and UEFI?

👉 **Answer:**
UEFI is an advanced version of BIOS with better speed, security, and support for large disks (>2TB).

❓ Q5: How to identify UEFI or Legacy?

👉 **Answer:**

- Mouse works → UEFI
- Only keyboard → Legacy
- Spinning circle boot → UEFI
- Straight line → Legacy

🔹 **BOOTING QUESTIONS**

❓ Q6: What is booting?

👉 **Answer:**
Booting is the process of starting a computer and loading the operating system into RAM.

❓ Q7: What is bootloader?

👉 **Answer:**
A program that loads OS:

- Windows → bootmgr
- Linux → GRUB

❓ Q8: What is boot device?

👉 **Answer:**
Device from which OS loads, like HDD, SSD, or USB.

🔹 **PARTITION QUESTIONS**

❓ **Q9: What is partitioning?**

👉 **Answer:**
Dividing a hard disk into multiple sections to organize OS and data.

❓ **Q10: What is MBR?**

👉 **Answer:**
Master Boot Record is an old partitioning system that supports up to 2TB disk and 4 partitions.

❓ **Q11: What is GPT?**

👉 **Answer:**
GUID Partition Table is a modern partitioning system that supports large disks and multiple partitions with backup.

🔹 **PARTITION TYPES QUESTIONS**

❓ **Q12: What is primary partition?**

👉 **Answer:**
Main partition where OS is installed.

❓ **Q13: What is extended partition?**

👉 **Answer:**
A container that holds logical partitions (only in MBR).

❓ Q14: What is logical partition?

👉 **Answer:**
Partitions inside extended partition used for storing data.

❓ Q15: What is active partition?

👉 **Answer:**
Partition that contains boot information and is used to start OS.

🔹 EFI & RECOVERY QUESTIONS

❓ Q16: What is EFI partition?

👉 **Answer:**
EFI System Partition contains boot files used by UEFI to start the OS.

❓ Q17: What is recovery partition?

👉 **Answer:**
A hidden partition used to repair, reset, or troubleshoot the system.

🔹 SECURE BOOT QUESTIONS

❓ Q18: What is Secure Boot?

👉 **Answer:**
A security feature in UEFI that prevents unauthorized or malicious software from loading during boot.

🔹 **TROUBLESHOOTING QUESTIONS (VERY IMPORTANT)**

💾 **BOOT ISSUE**

❓ **Q19: No Bootable Device Found?**

👉 **Answer:**

Possible causes:

- Wrong boot order
- OS missing
- Disk failure
- Bootloader corrupt

👉 Solution:

- Check BIOS boot order
- Repair bootloader
- Check disk

🔌 **USB NOT DETECT**

❓ **Q20: USB not booting?**

👉 **Answer:**

- Enable USB boot in BIOS
- Check bootable USB (Rufus)
- Change boot priority

⚠️ OS INSTALL ISSUE

❓ Q21: OS not installing?

👉 **Answer:**

- Partition type mismatch (MBR vs GPT)
- Secure boot issue
- Corrupt ISO

🔹 REAL JOB SCENARIOS

❓ Q22: System stuck on logo?

👉 **Answer:**

- Bootloader issue
- OS corruption
  👉 Solution:
- Use recovery mode
- Repair startup

❓ Q23: BIOS settings resetting?

👉 **Answer:**
👉 CMOS battery dead

❓ **Q24: Laptop not booting after update?**

👉 **Answer:**
👉 Boot files corrupted
👉 Use recovery or reinstall OS

🔹 **INTERVIEW IMPRESSION QUESTIONS**

❓ **Q25: What is full boot flow?**

👉 **Answer:**
BIOS → POST → Bootloader → OS → Login screen

❓ **Q26: How do you troubleshoot boot issue?**

👉 **Answer:**

- Check BIOS
- Check boot order
- Check disk
- Repair OS
